EVERY BOUNDED COHERENT COMPLEX ON AN OPEN EXTENDS TO A BOUNDED COHERENT COMPLEX ON THE AMBIENT NOETHERIAN SCHEME

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Editorial note. Input proposition-DCoh is same-chapter NOT carded (cross-line-claimed label) and is declared external.

Allowed standard prerequisites (editorial).

- Bounded coherent complexes are represented by bounded complexes of coherent sheaves. proposition-DCoh (same chapter, NOT carded, declared external). Represents K by a bounded complex F-bullet of coherent O_U-modules.
- Quasi-coherence of pushforwards along quasi-compact separated morphisms. Schemes, lemma push-forward-quasi-coherent (cross-chapter, named). The terms of j_*F-bullet are quasi-coherent on X.
- Extending coherent subsheaves from opens. Properties, lemma extend (cross-chapter, named). Provides coherent submodules of j_*F^i restricting to the given coherent sheaf on U.

Source reference. Original reference 01PF: focused-seeds/geometry/S9-perf2-lift-coherent/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 0FDB: sources/stacks-perfect/pinned-original-source.tex, source lines 2813–2813. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01LC: focused-seeds/geometry/S9-perf2-lift-coherent/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a Noetherian scheme and let $j : U \to X$ be an open immersion. The functor $D^b_{Coh}(\mathcal{O}_X) \to D^b_{Coh}(\mathcal{O}_U)$ is essentially surjective.*

Proof. Let K be an object of $D^b_{Coh}(\mathcal{O}_U)$. By Proposition 0FDB we can represent K by a bounded complex $\mathcal{F}^\bullet$ of coherent $\mathcal{O}_U$-modules. Say $\mathcal{F}^i = 0$ for $i \notin [a, b]$ for some $a \leq b$. Since j is quasi-compact and separated, the terms of the bounded

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complex $j_*\mathcal{F}^\bullet$ are quasi-coherent modules on X, see Schemes, Lemma 01LC. We inductively pick a coherent submodule $\mathcal{G}^i \subset j_*\mathcal{F}^i$ as follows. For $i = a$ we pick any coherent submodule $\mathcal{G}^a \subset j_*\mathcal{F}^a$ whose restriction to U is $\mathcal{F}^a$. This is possible by Properties, Lemma 01PF. For $i > a$ we first pick any coherent submodule $\mathcal{H}^i \subset j_*\mathcal{F}^i$ whose restriction to U is $\mathcal{F}^i$ and then we set $\mathcal{G}^i = \mathrm{Im}(\mathcal{H}^i \oplus \mathcal{G}^{i-1} \to j_*\mathcal{F}^i)$. It is clear that $\mathcal{G}^\bullet \subset j_*\mathcal{F}^\bullet$ is a bounded complex of coherent $\mathcal{O}_X$-modules whose restriction to U is $\mathcal{F}^\bullet$ as desired. $\square$